MASTER'S THESIS DEFENSE · BIOMEDICAL ENGINEERING

Development of a Clinical PET/CT Imaging Dataset for AI-Based Cancer Detection

A curated 124-patient ^{18}F-FDG PET cohort in head-and-neck cancer, validated with a leakage-safe, site-aware protocol

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OUTLINE

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BACKGROUND

PET/CT: the reference whole-body cancer modality

20–30%

higher diagnostic accuracy for PET+CT than CT alone [4]

17–21%

SUVmean / SUVmax differences between unharmonized scanners [9]

SUVbw

decay-corrected uptake per injected dose and body weight; basis of EARL accreditation [5], [6]

^{18}F-FDG maps glucose metabolism (the Warburg effect). That **measurability makes PET tractable for AI — and fragile**, because uptake values shift with scanner and protocol [1]–[6].

THE DATA BOTTLENECK

Models are mature; the data are not

1 · Acquisition heterogeneity

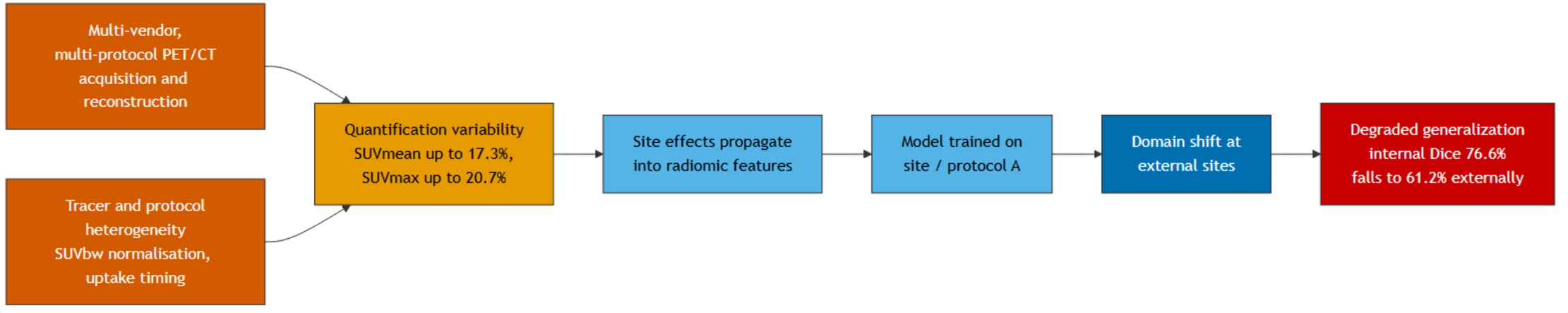
Models can learn a scanner signature instead of a tumour phenotype; harmonisation (EARL, ComBat) is a design requirement [5], [6], [9], [10]

2 · Annotation

Ground truth is a contour; only ~55% of studies use multi-expert labels [7], [8], [14]

3 · Study design

Patient-level leakage and site-skewed splits inflate performance [15], [16]



PROBLEM STATEMENT

A real clinical cohort with four hard properties

0 CT
Labelled PET/CT, but no CT series was exported — a PET-only study

10 / 54
Scanners and reconstruction geometries across 124 patients

12
CERR re-exports with no dosimetry — SUV must be imputed

8 / 18
Local recurrences and deaths — very few events to learn from

The question is not only **“how well does a model score?”** but **“how much of that score survives** when site, leakage and chance are controlled?”

RESEARCH QUESTIONS AND OBJECTIVES

Four questions, six objectives

RQ1 Can a heterogeneous clinical ^{18}F-FDG PET collection be made analysis-ready — audited, calibrated and annotation-validated — when metadata are inconsistent or absent?

RQ2 Do handcrafted metabolic and shape/texture features of the primary tumour carry cross-validated prognostic signal for local control?

RQ3 Does that signal survive nested CV, a within-site permutation null and orthogonalisation against clinical covariates?

RQ4 What do the results imply for endpoint choice, event counts and reporting in small multi-scanner cohorts?

Objectives

1. Assemble and audit the cohort
2. Robust preprocessing and SUV quantification
3. Clinical, metabolic and shape/texture blocks
4. Leakage-safe, site-aware validation with chance calibration
5. Quantify signal for local control and vital status
6. Dataset and reporting recommendations

LITERATURE REVIEW

Reported AI performance in PET oncology

Approach / benchmark	Reported result
Handcrafted radiomics (226-study review)	AUC 0.811
Deep radiomics	AUC 0.842
End-to-end deep learning	AUC 0.846
Hybrid radiomics + DL	AUC 0.861
HECKTOR (H&N segmentation / outcome)	Dice ≈ 0.78 · C-index 0.723
nnU-Net, NSCLC PET/CT	Dice 83.4 internal · 81.1 external

59% of studies use small cohorts

79% rely on private data

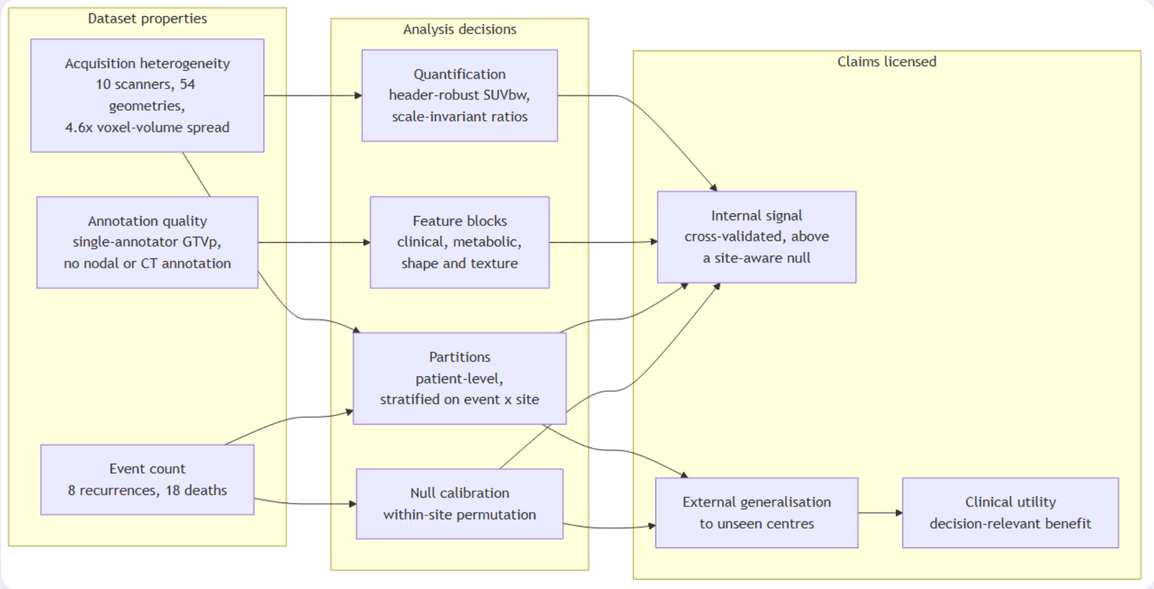
~48% adhere to IBSI feature standards

Headline numbers cluster at 0.81–0.86 — but few are tested **across sites** or **against a chance baseline** [4], [8], [17].

RESEARCH GAPS

Six gaps that cascade into what a study can claim

1. **Small, single-centre cohorts** — 59% small, 79% private [8]
2. **Weak external validation** — Dice drops 21% under domain shift [4], [28]
3. **Incomplete harmonisation** — ComBat needs 20–30 patients per batch [9], [10]
4. **Annotation and documentation** — 55% multi-expert, ~48% IBSI [8], [17]
5. **Site confounding unmeasured** — separability rarely reported
6. **Reporting gaps** — no fold coverage, no structured null [18]



METHODOLOGY

From raw DICOM to a defensible estimate

1

Cohort audit

Integrity, geometry, units and endpoints of 124 patients; 10 scanners, 54 geometries

→

2

Quantification

Header-robust SUVbw with site calibration; imputation for 12 CERR series; annotation checks

→

3

Feature blocks

Clinical 7, metabolic 15, shape/texture 11 → combined 33, on an isotropic grid

→

4

Validation

Four stages: naive CV → coverage → site-aware with nulls → nested CV

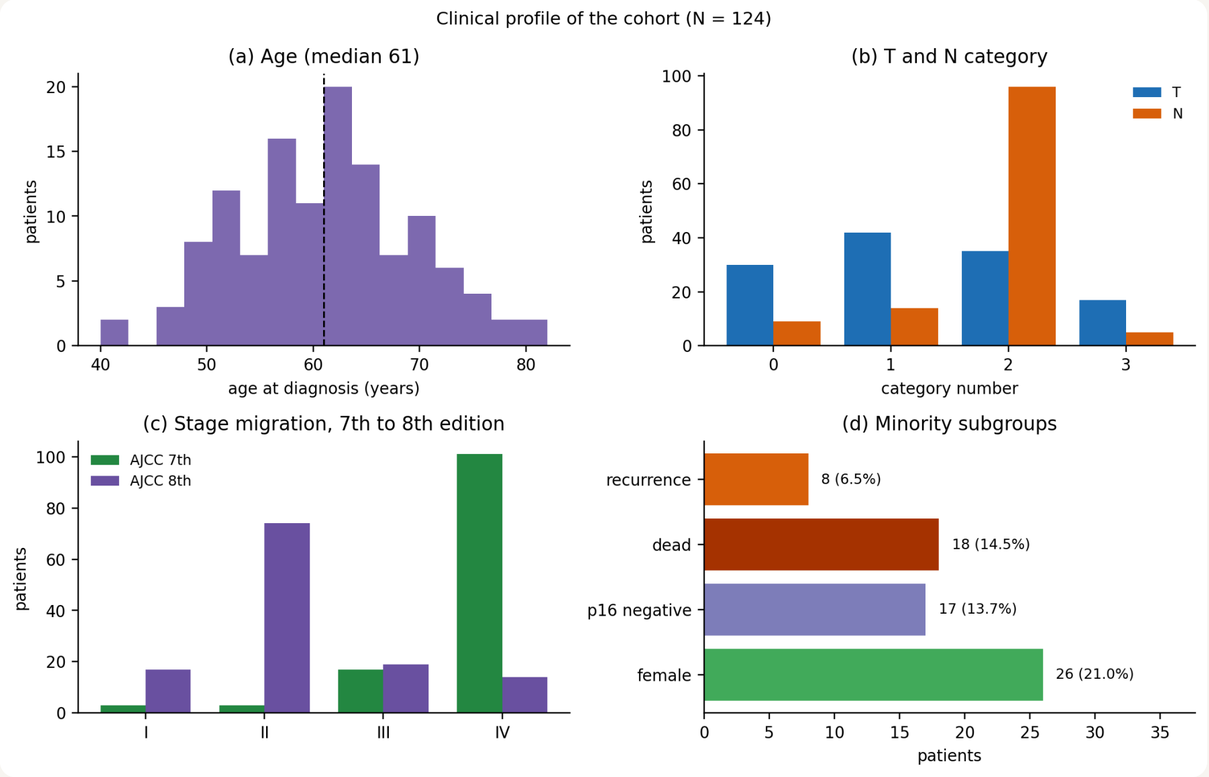
Endpoints: local control (8 events) and vital status (18 deaths)

Rule: all preprocessing fitted inside each training fold (sklearn Pipeline)

DATASET · COHORT

124 HNSCC patients with ^{18}F-FDG PET and a GTVp contour

Median age (IQR)	**61** (55–66)
Male	**79%**
p16-positive	**86.3%**
Local recurrence	**8** (6.5%)
Deaths	**18** (14.5%)
Both events	**3**
Median GTVp volume	**14.65 mL** (8.22–26.07)
Missing clinical values	**0**



DATASET · ACQUISITION HETEROGENEITY

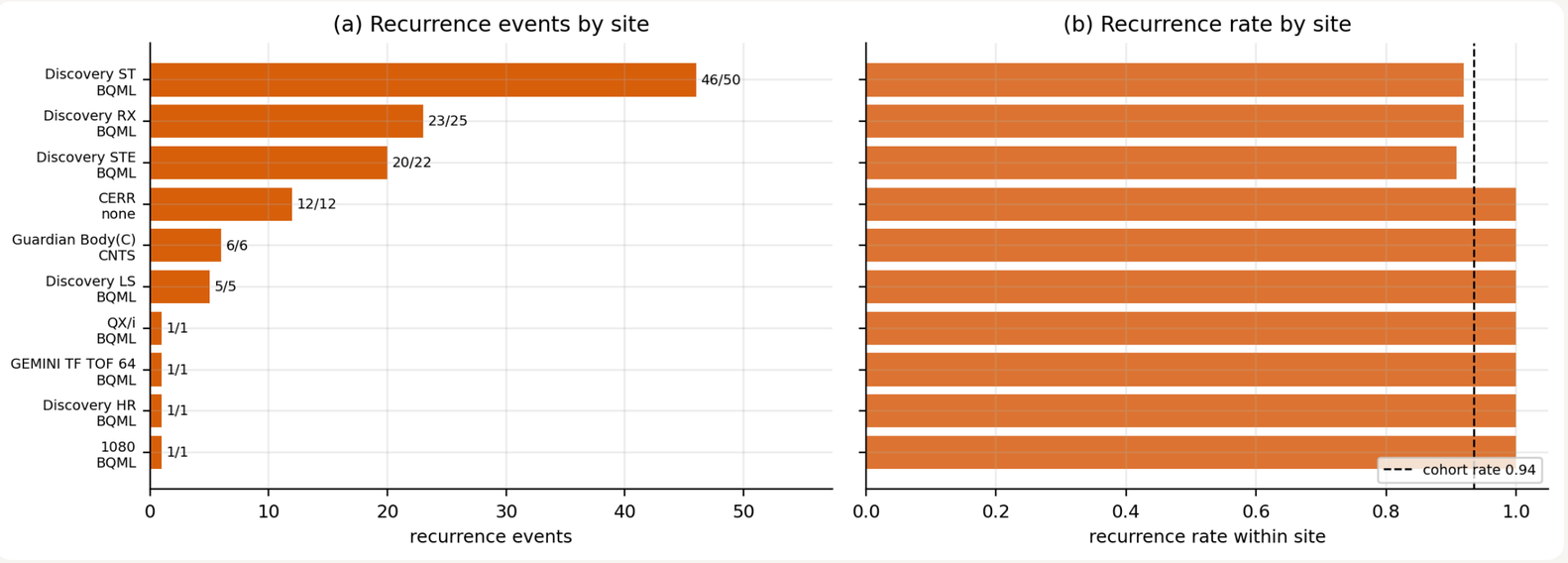
Ten scanners, and events concentrated in three

54
distinct acquisition geometries

4.6×
voxel volume spread (0.021–0.098 mL)

2.0×
in-plane spacing spread (2.67–5.47 mm)

106 / 6 / 12
pixel units BQML / CNTS / absent



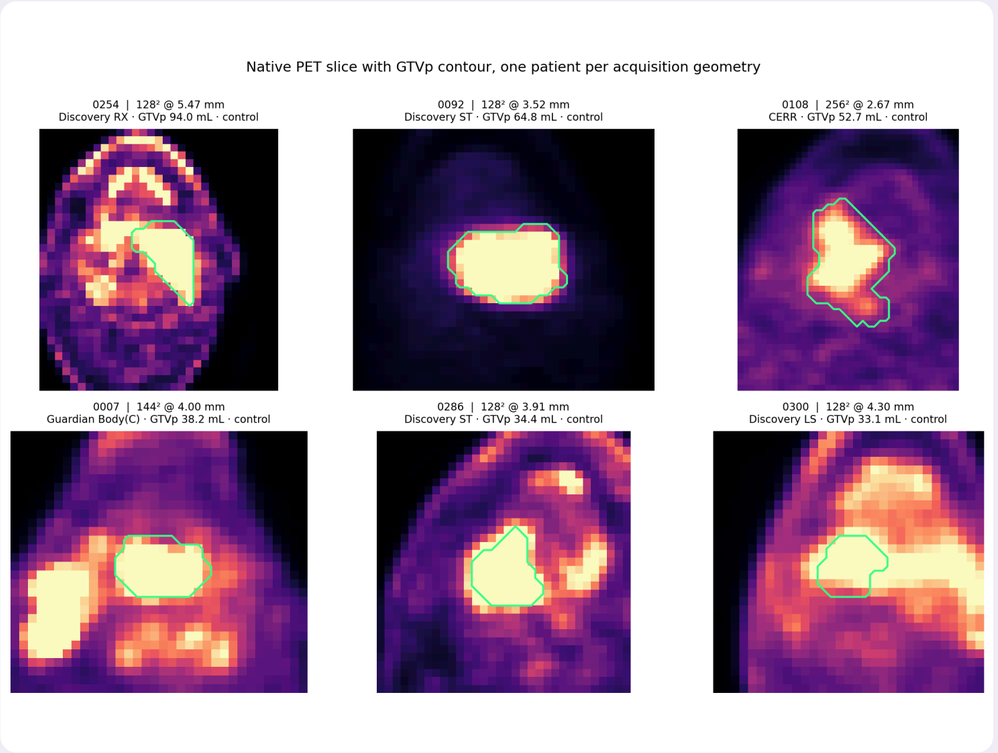
All **8 recurrences** come from **3 scanners** (Discovery ST 4, RX 2, STE 2) — site and outcome are entangled.

Header-robust SUVbw calibration

$$SUV = k_{site} \cdot pixel \cdot W / (D \cdot f_{decay})$$

$$f_{decay} = 2^{-\Delta t / T½}$$

- Per-site factor k_{site}; DoseCalibrationFactor rejected as a converter
- 12 CERR series imputed: 64 kg, 370 MBq, decay 0.5
- **124/124** GTVp contours passed the physiological check
- Median tumour SUVmax **127.6** within the plausible 80–180 range



FEATURES AND LEARNERS

A feature ladder: clinical → imaging → combined

7

Clinical — age, sex, p16 and ordinal staging; zero missing

15

Metabolic — absolute SUV descriptors plus scale-invariant tumour-to-body ratios

11

Shape / texture — 2 mm isotropic shape, first-order and GLCM descriptors

33

Combined — all blocks concatenated

L2 logistic regression

Regularised linear baseline, well suited to few events

Shallow random forest

200 trees, depth 3 — non-linear but constrained

28-configuration grid

4 feature sets × (3 C × 2 class weights) for LR + 1 RF each; inner folds only

Shape is computed on a **2 mm isotropic grid** — with a 4.6× voxel-volume spread, native-grid sphericity would describe the scanner, not the tumour.

VALIDATION DESIGN

Four stages, each removing one source of optimism

STAGE 1

Naive stratified CV

5-fold, outcome-stratified. Shows what a typical study would report.

STAGE 2

Site-shift coverage

Leave-one-site / geometry-out and random site hold-outs. How many folds are even scoreable?

STAGE 3

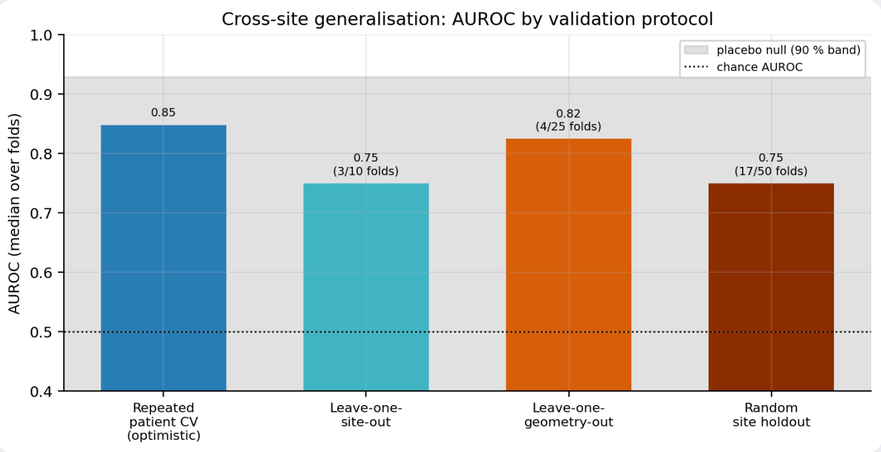
Site-aware CV + nulls

Folds stratified on site × outcome, 20 repeats, within-site permutation null, deconfounding.

STAGE 4

Nested CV

10 × 5 outer folds; model selection inside. The primary, selection-free estimate.

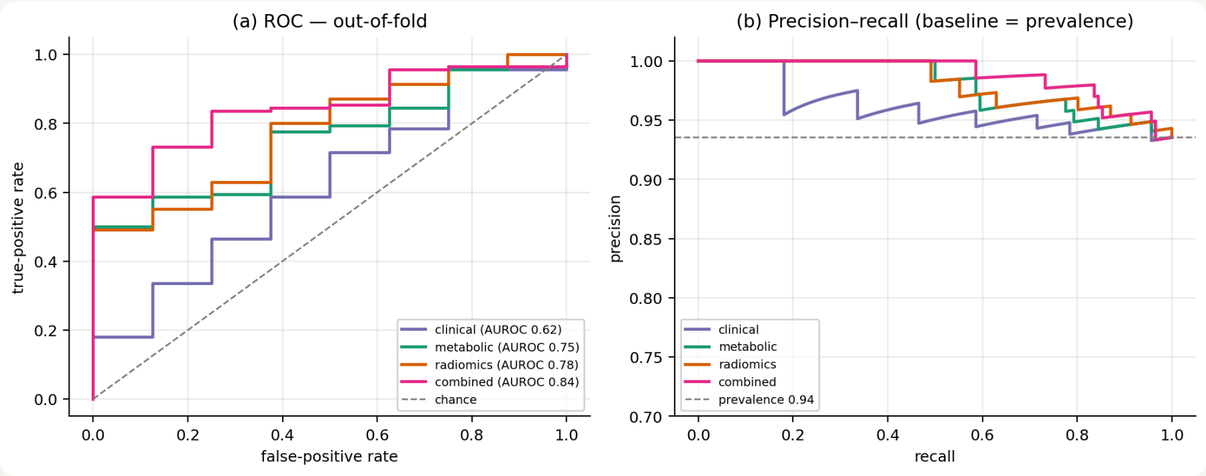


Every stage reports **fold coverage** and an **explicit chance level**, so the reader can see how much of a score is signal.

STAGE 1 · NAIVE CROSS-VALIDATION

The AUPRC trap: 0.96–0.99 is almost chance

Feature set · learner	AUPRC	AUROC
Combined · LR	**0.988**	**0.843**
Radiomics · RF	0.986	0.836
Radiomics · LR	0.981	0.778
Metabolic · LR	0.980	0.752
Combined · RF	0.977	0.768
Metabolic · RF	0.971	0.713
Clinical · RF	0.964	0.666
Clinical · LR	0.961	0.623

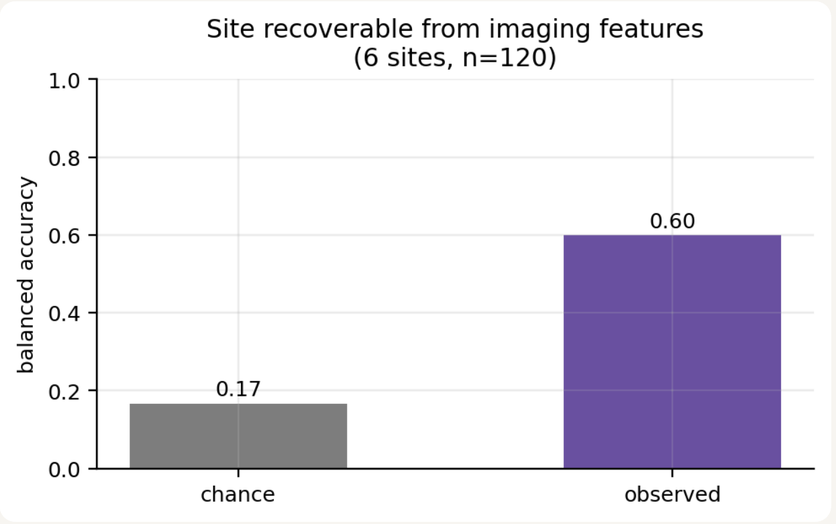


With 93.5% of patients controlled, a **constant predictor scores AUPRC ≈ 0.94**. AUROC (0.62–0.84) is the informative metric.

STAGE 2 · SITE-SHIFT COVERAGE

Most site hold-out folds cannot be scored

Protocol	Scoreable folds	AUROC mean / median
In-distribution	50/50 (100%)	0.840 / 0.848
Leave-one-site-out	3/10 (30%)	0.755 / 0.750
Leave-one-geometry-out	4/25 (16%)	0.841 / 0.824
Random site hold-out	17/50 (34%)	0.748 / 0.750



0.600

site predicted from features (balanced accuracy, SD 0.097) vs chance 0.167 — 6 sites, 120 patients

0.00–0.93

naive null AUROC across these folds (mean 0.528) — single folds are uninterpretable

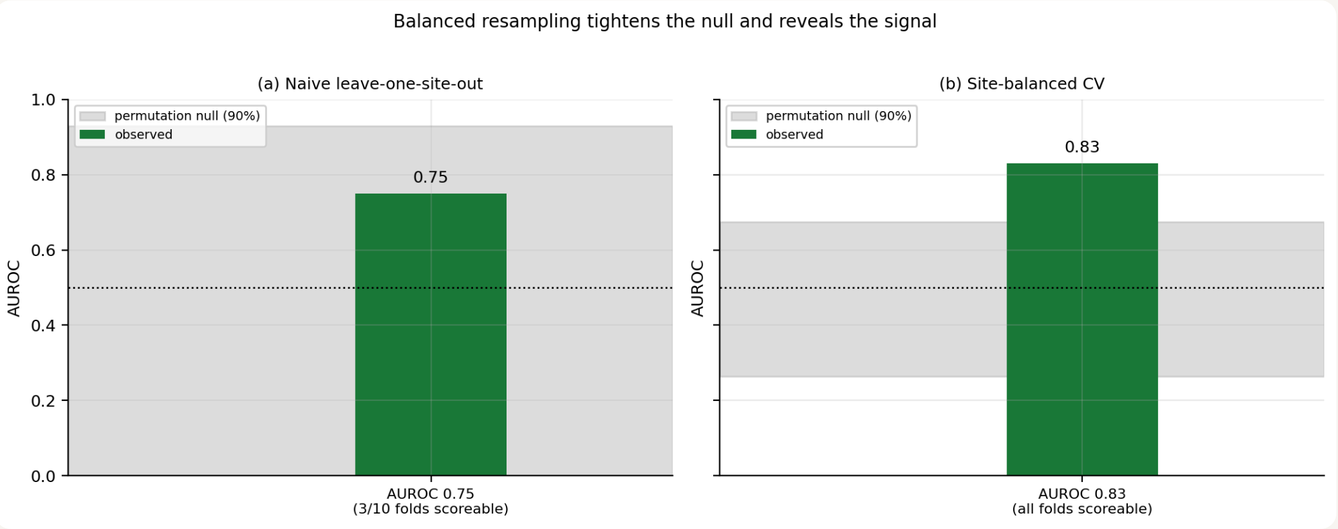
STAGE 3 · SITE-AWARE CV WITH PERMUTATION NULLS

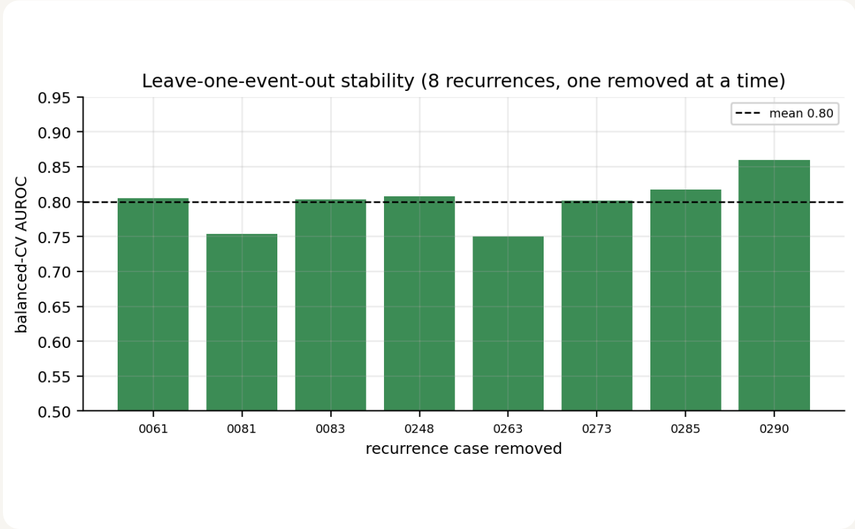
The signal sits above a within-site null

0.836 ± 0.024
pooled AUROC (event-matched 0.830 ± 0.028), 20 repeats

0.263–0.674
within-site null 90% band (mean 0.480)

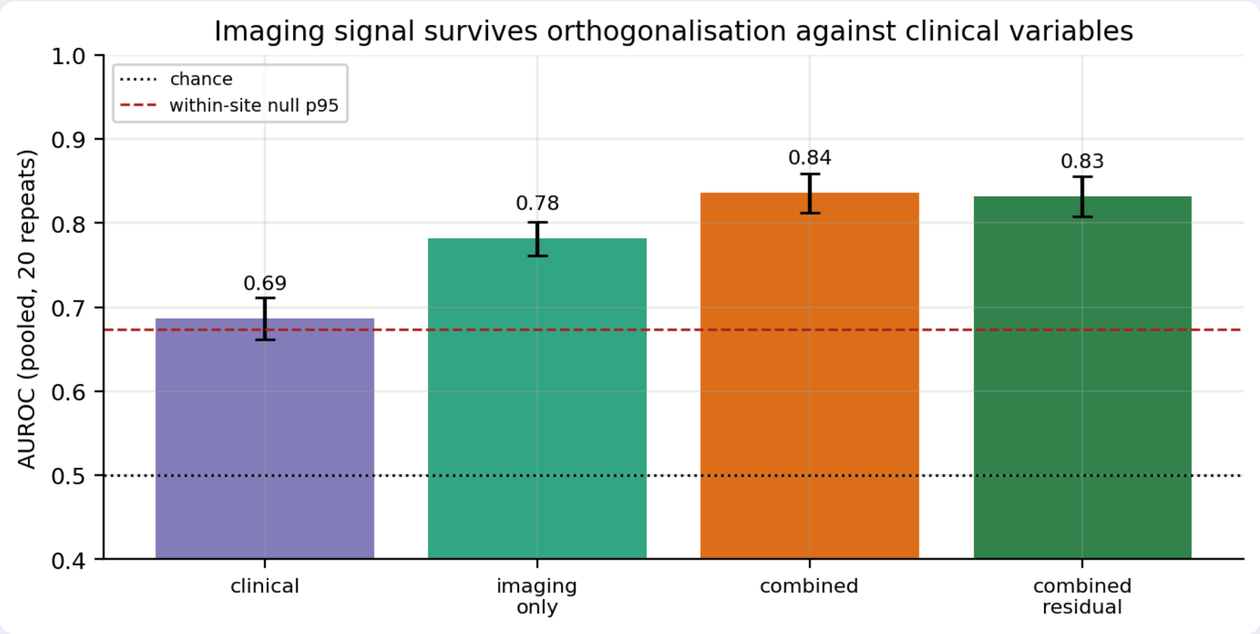
0.750–0.860
leave-one-event-out AUROC (mean 0.800)





STAGE 3 · DECONFOUNDING

Imaging adds signal beyond clinical covariates



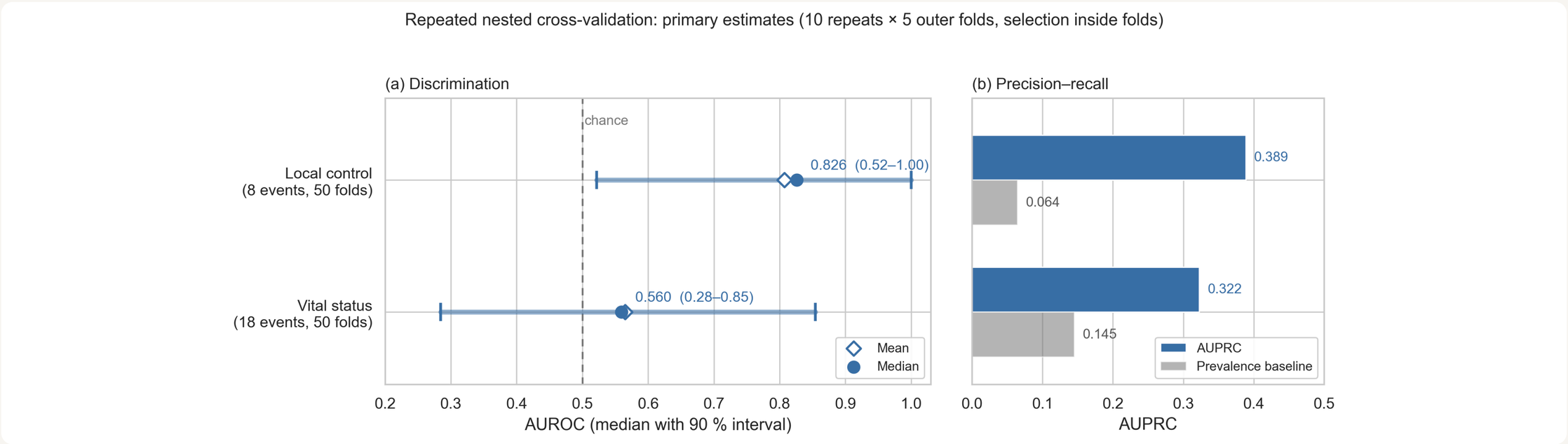
Clinical only	0.687
Imaging only	0.781
Combined	0.836
Imaging residualised on clinical	0.832

+0.145 from adding imaging; only **−0.004** after orthogonalising against clinical variables.

STAGE 4 · NESTED CV — PRIMARY ESTIMATE

Local control holds; vital status does not

Endpoint	Median AUROC (mean)	90% interval	AUPRC vs baseline	Most-selected model
Local control	**0.826 (0.807)**	0.522–1.000	0.389 vs 0.064	Combined · LR (29/50)
Vital status	**0.560 (0.565)**	0.284–0.854	0.322 vs 0.145	Clinical · LR (19/50)



STAGE 4 · NESTED NULLS AND MODEL SELECTION

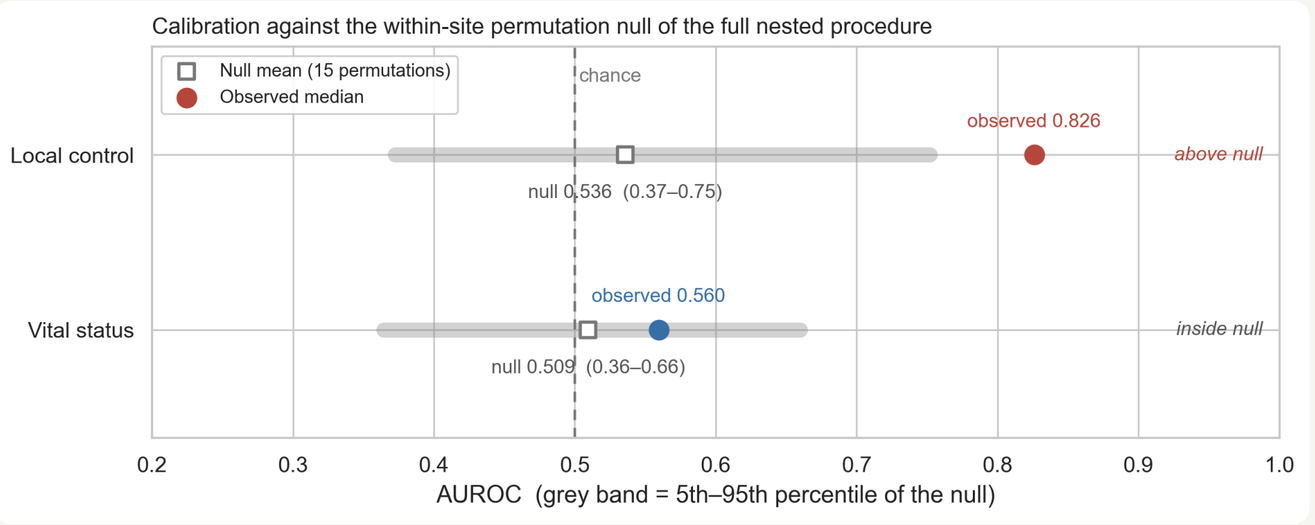
Only local control clears the nested null

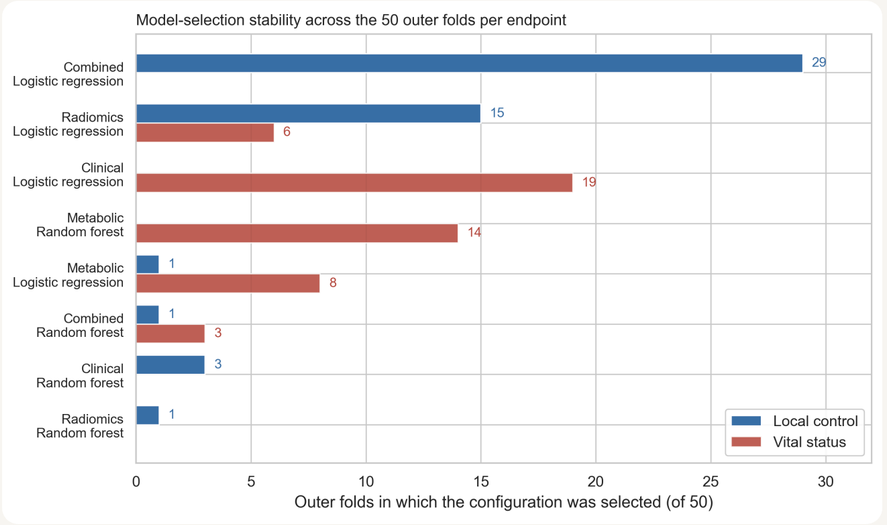
0.373–0.752

Local control null band (mean 0.536) — observed 0.826 lies **above**

0.364–0.660

Vital status null band (mean 0.509) — observed 0.560 lies **inside**





CONCLUSION

The dataset and validation design, not the model, set the ceiling on the claim

Contributions

- Curated, audited 124-patient FDG PET dataset
- Header-robust SUVbw calibration
- Clinical → imaging → combined feature ladder
- Four-stage, site-aware validation protocol
- Local-control signal (0.826) and a negative vital-status result
- Reporting recommendations: coverage and chance

Limitations

- Only 8 local-control events
- No CT series; PET-only
- Single reader, GTVp only
- 10 scanners; site separable from features
- 12 patients with imputed dosimetry
- Internal validation; handcrafted features; coarse nulls

Future work

- External, pre-registered validation
- Acquire the CT series
- Multi-annotator ground truth and nodal disease
- Harmonisation-first re-extraction
- Event-rich pooled cohorts
- Release pipeline with a datasheet

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